

maritime plants 63
 marlins 57
 Mauna Loa 22
 melting ice 86
 meltwater 33, 86, 93
 methane 73, 75, 86–87, 91–93
 microbe 93
 Micronesia 83
 Mid-Atlantic Ridge 20–21, 23, 90–91
 midocean ridges 13, 16, 20–21, 72, 75, 91–93
 Middle East 74
 Mississippi Delta 85
 mobile starfish 60
 mollusks 56, 93
 monsoon 50–51, 93
 monsoon currents 50
 Mount Pinatubo 31
 mudflats 28–29, 63
 mudflows 30
 mussels 58, 60, 62, 73

N

natural antifreezes 65
 natural breakwaters 45
 navigators 10–11
 neap tide 46, 62, 93
 Newfoundland 88
 New Guinea 51
 new Moon 46
 New Orleans 43
 no fishing zones 88
 North America 61, 91
 North Atlantic 42, 48, 91
 North Atlantic Deep Water 52–53
 North Atlantic gyre 48, 49
 North Pacific gyre 48
 North Sea 81
 Northwater 64
 nuclear submarine 15

O

ocean-floor map 17, 20, 91
 ocean basins 18
 ocean conveyor 53
 ocean current disruption 87
 ocean currents 13, 36, 54, 69, 77, 86–87, 91
 ocean depths 70–71
 ocean floor 12, 16–17, 20–21, 23–26, 31–32, 68, 73–75, 91
 ocean floor sediments 30–31, 91
 ocean formation 18–19
 oceanic boundary 24
 oceanic crust 18, 20, 22, 24, 93
 oceanic rocks 21
 oceanographic institutions 12
 oceanography 12–13, 91
 oceans
 and continents 18–19
 and redistribution of water 48–49
 and the atmosphere 40–41
 as a source of energy 76–77
 carbon dioxide in 86, 87
 chemistry 35
 density 52
 dissolved minerals in 54
 environments 9
 exploration 9
 habitat destruction in 84–85

heat distribution in 36–37
 ice cover 38–39
 ice formation 38
 marine life 38–39, 49–51, 54–68, 70–71, 80–81, 84, 86–88
 mineral resources 74–75
 natural resources 9
 nutrients 54–55, 68–69
 on other planets 8
 pressure 14–15, 37, 43, 59
 research see oceanography
 size and depth 9
 sound 37
 temperature 36–37
 topography 16–17
 trade and tourism 82–83
 underwater life 58–59
 volume 8
 water in 34–37, 52–53
 ocean trenches 16, 24, 93
 octopus 59
 oil and gas 74, 86
 oil spill 84, 91
 Oparin, Aleksander 90
 orange roughy 80
 overfishing 80–81, 84, 89
 oyster shells 75

P

Pacific islands 10
 Pacific Ocean 9–10, 14, 16, 22, 25, 45, 49, 50–51, 53, 55, 78, 82, 90–91
 Pacific Ring of Fire 25
 pack ice 38, 52, 64–65
 Panama Canal 82, 90
 pancake ice 38
 Pangaea 19, 90
 Panthalassa 19
 pearls 75
 penguins 56–57, 64–65
 peridotite 18
 Persian Gulf 74
 Peru 51, 78
 Peru-Chile Trench 24
 pesticides 84
 photophores 70, 93
 photosynthesis 54, 93
 phytoplankton 54–57, 60, 66, 70, 93
 Piccard, Jacques 14
 pillow lavas 21, 93
 plankton 36–37, 49, 64, 68–69, 79, 84, 93
 plankton blooms 54, 68, 93
 plankton distribution 13
 plankton swarms 56
 plastic waste 84
 plunging breakers 45
 polar bear 65, 86
 polar cell 40
 polar easterlies 41
 polar ice 52–53
 polar oceans 9, 38–39, 52–53, 61, 64–65, 86
 polar regions 53
 polar winters 38–39
 pollution 84–87, 89
 polynas 64
 Polynesians 10, 90
 Pompeii worm 72
 ports 82
 prawns 59
 pressure-proof suits 37
 prevailing winds 40–42, 93
 purse seine nets 78, 80

R

racers 47
 radiolarians 56
 rain 40, 42–43
 rat-tail fish 71
 rays 60
 receding tide 63
 red coralline algae 66
 Red Sea 83, 90
 red tide 84
 reef 93
 reef fish 85
 reef shark 67
 reef zones 66
 rift 93
 rifting 20
 rift valley 20
 Roaring Forties 41
 rock arch 28
 rocky shores 63
 rogue waves 45
 Ronne Ice Shelf 52
 Ross Ice Shelf 39, 90
 Ross, James Clark 11, 90
 Ross Sea 52, 90
 ROVs 15, 93

S

salmon farming 79
 salps 56
 salt 35
 salt marshes 63, 93
 Saltstraumen tidal race, Norway 47
 sandstone 31
 Sargasso Sea 49
 satellite images 13, 31, 42–43, 49, 55, 60, 64, 68, 79
 Saudi Arabia 74
 scallops 59
 Scandinavia 32
 Scripps Institution 12, 91
 scuba diving 83, 90, 93
 seafloor sediments 54
 sea anemones 58, 60, 62, 66
 seabed 30, 60, 86
 seabirds 51, 56, 60–61, 64, 80–81
 sea cucumber 71
 seafloor 32, 59, 74–75
 seagrass 55, 66, 85
 seagrass meadows 63
 seahorse 63
 sea ice 38, 39
 sea level 8, 32–33, 86
 seals 60, 64–65, 80
 seamounts 13, 22–23, 69, 93
 sea otters 60, 61
 sea salt 74
 sea slugs 59, 67
 sea snails 59
 seasonal wind shifts 50–51
 sea urchins 59, 60–61, 67
 seaweed farms 79
 seaweed 55, 62
 sedimentary rocks 30–32, 93
 sewage treatment 89
 Shackleton, Ernest 38, 90
 shale 31
 shallow seas 60–61
 Shanghai 86
 sharks 57, 59, 69, 81
 shellfish 65, 67
 shingle banks 28

Shinkai 6500 15
 shoaling fish 57, 68
 shoals 56
 shock waves 26
 shorebirds 63
 shoreline 46
 shrimp 59, 67, 71, 73
 shrimp pools 78–79
 sodium chloride 74
 SOFAR 37
 Somali Current 50
 sonar 13, 90–93
 sonar image 21
 sonar surveys 13
 sounding 12, 93
 South America 51, 82
 South Equatorial Current 51
 Southern Ocean 9, 11, 20, 45, 52, 57, 64, 78, 90–91
 southwest Africa 75
 Spain 54
 spawn 93
 sperm whale 59
 spits 29
 spring tide 46, 62, 93
 squid 59, 65, 70
 stacks 28
 starfish 59, 67
 St. Malo, France 77
 storm clouds 43
 storm surge 43, 90–93
 strait 93
 stratosphere 40
 subduction zones 24–26, 93
 submarine canyon 30
 submarine chasm 24
 submarine earthquakes 20
 submarine fans 30, 93
 submarine geysers 72
 submarine mountains 21
 submarine volcanoes 69
 submerged seamounts 22
 submerged volcanoes 22–23
 submersibles 14–15, 91, 93
 subtropics 40
 Suez Canal 82, 90
 Sumatra 27
 Sunda Arc 24
 sunlit zone 60, 70, 93
 supercontinent 19, 93
 superheated water 23, 93
 surface current gyres 53
 surface currents 48–53
 swell 44, 93
 swordfish 57

T

Tahiti 83
 Tambora volcano 31
 technology 78–79, 80, 88
 temperate regions 42
 Sharp, Marie 17, 91
 thermocline 36, 93
 thermohaline circulation 52, 93
 tidal barrages 77, 91
 tidal bulge 46
 tidal cycles 46
 tidal pools 79
 tidal race 93
 tidal range 93
 tidal shores 62–63
 tidal streams 46–47, 93
 tides 46–47, 62–63, 76–77, 86
 tides, low and high 47
 Tierra del Fuego 12
 tiger prawns 79
 Timor Sea 10

Titanic 14, 90–91
 titanium 75
 tornado 43
 tourism 82–83
 toxins 85
 trade winds 41, 51, 93
 trading routes 82
 transform boundary 19, 93
 transform faults 20
 trawling 12
 Trieste 91
 tropical oceans 36, 66–68
 tropical seas 9
 tropical winds 40
 tropics 42
 troughs 26
 tsunamis 19, 24, 26–27, 93
 tuna 57, 78, 81
 turbidity current 30, 93
 turtles 80
 twilight zone 70, 93
 typhoons 42

UV

undersea volcanoes 16–17
 untreated sewage 84–85, 89
 uplift 20
 upwelling zones 49, 61, 69, 93
 USS *Nautilus* 91
 Vikings 10, 90
 volcanic cataclysms 25
 volcanic chains 22
 volcanic islands 12, 22–23

W

Walsh, Don 14
 water cycle 34–35
 waterspout 43
 water vapor 34, 40, 43, 93
 wave development 44
 wave energy 28, 44–45, 76
 wavelength 44–45, 93
 wave refraction 45
 waves 26, 28–29, 44–45
 weather systems 40
 Weddell Sea 38, 41, 52, 90–91
 weddell seals 65
 Wegener, Alfred 90
 west African coast 60
 westerlies 41
 whales 56, 64
 whale shark 57, 68
 whaling 81
 whirlpools 47
 wildlife hotspots 69
 wild salmon 79
 Wilkes, Charles 11
 wind farms 76
 wind power 76
 winds 44–45, 48–53, 76–77
 wind turbines 76
 Woods Hole Oceanographic Institution 12, 14, 91
 world fish stocks 80, 88
 worms 60, 63

YZ

Yucatan 33
 Zanzibar, east Africa 79
 zooplankton 56, 70, 93
 zooxanthellae 66, 93